



QuietChat: No Judgement, Just Match Vibes

v1.0
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1. APP ENTRY & USER ACQUISITION FLOW

1.1 Objective

Drive user conversion from install → signup → first meaningful action (match/chat) with minimal friction while ensuring fraud prevention.

1.2 Detailed User Journey Flow

1. App Install
2. Splash Screen Initialization
3. Login Screen
4. OTP Verification
5. Profile Setup
6. Intent Selection
7. Gamified Onboarding
8. Profile Score Generation
9. First Match Unlock

1.3 Splash Screen (System Initialization Layer)

UX

- App logo + tagline
- 1.5–2 sec duration

Backend Processes

- Fetch remote config (Firebase Remote Config)
- Load feature flags (A/B experiments)
- Initialize SDKs:
 - Analytics (Firebase / Mixpanel)
 - Crash reporting
- Pre-fetch minimal user metadata (if returning user)

Edge Cases

- Slow network → fallback cached config
- App cold start failure → retry logic

1.4 Login Options

Primary: Mobile OTP (India optimized)

UX Flow

- Enter mobile number
- OTP auto-detect (Android)
- Timer-based resend

Backend

- OTP service (Twilio / MSG91)
- Rate limiting (max 5 attempts/hour)
- Device + number mapping



Secondary: Google / Apple Login

- OAuth token verification
- Map social ID to internal user ID

Fraud Prevention

- Same device + multiple accounts → flagged
- Suspicious IP → restrict access

2. AUTHENTICATION & USER CREATION

2.1 OTP Verification Logic

- OTP expiry: 2 minutes
- Retry cooldown: 30 sec
- Max attempts: 5

2.2 Account Creation (Backend)

Objects Created

- User Table Entry
- Wallet Object
- Empty Profile Record
- Analytics Event (Signup)

Schema Example

User:

- user_id
- phone
- created_at
- device_id

Wallet:

- credits
- tokens
- audio_balance

2.3 Free Credits (Activation Engine)

Allocation

- 5 Match Spins
- 20 Messages
- 5 Audio Minutes

Note: It is configurable from the admin panel

Behavioral Strategy

- Push user to first match



- Encourage chat initiation
- Create habit loop

Abuse Control

- One-time per device
- Linked with phone + device fingerprint

3. PROFILE SETUP (DATA FOUNDATION – DETAILED)

3.1 Objective

To collect **structured + behavioral + emotional data** that powers:

- Matching algorithm accuracy
- Personalization
- Trust & safety
- Engagement and retention

This section defines **what data to collect, how to collect it, and how to use it.**

3.2 PROFILE CREATION FLOW (STEP-BY-STEP UX)

Step 1: Basic Info

→ Name → Age → Gender → Location

Step 2: Profile Photos

→ Upload 1–6 images

Step 3: Interests & Preferences

→ Select tags

Step 4: Bio & Prompts

→ Add personality

Step 5: Advanced Inputs

→ Voice intro + quiz

Step 6: Verification (Optional but encouraged)

Step 7: Completion Score Display

3.3 CORE PROFILE DATA MODEL

3.3.1 Basic Identity (Mandatory)

Field	Type	Validation	Purpose
Name	String	2–30 chars	Identity
Age	Integer	≥18	Legal + matching
Gender	Enum	M/F/Other	Matching filter
Location	Geo	GPS/manual	Proximity

3.3.2 Profile Photos (Critical for Engagement)

Rules

- Min: 1 photo
- Max: 6 photos

Features

- Crop + filter tools
- Auto face detection



- Photo order prioritization

AI Moderation

- NSFW detection
- Fake image detection
- Multiple face detection

Backend

- Store in S3
- CDN delivery

Scoring Impact

- +15–20% profile score

3.3.3 Bio (Free Text)

Input

- Max 300 characters

Suggestions

- Prompt-based templates
 - “My vibe is...”
 - “Looking for...”

NLP Usage

- Extract keywords
- Detect sentiment

Impact

- Improves match relevance
- Helps conversation starters

3.3.4 Interests (Tag-Based System)

Categories

Lifestyle

- Travel, Fitness, Foodie, Music

Personality

- Introvert, Extrovert, Funny

Activities

- Movies, Gaming, Reading

UX

- Multi-select chips
- Max 10–15 tags

Backend

- Stored as array
- Used in similarity scoring

3.3.5 Preferences (Advanced Matching Filters)

User Preferences

Field	Example
Preferred Gender	Female
Age Range	21–30
Distance	10–50 km
Intent Match	Same intent

Role

- Hard filter in matching



3.4 ADVANCED PROFILE DATA (HIGH VALUE)

3.4.1 Voice Introduction (Unique to Quietchat)

Input

- Max 30 seconds

UX

- Record → preview → upload

Benefits

- Builds trust
- Adds emotional layer

Backend

- Audio compression
- Storage + streaming

Scoring

- +20% weight

3.4.2 Personality Quiz

Structure

- 5–10 questions

Output

- Personality type
- Compatibility vector

Example

- “Weekend plan?”
- “Introvert vs Extrovert?”

Usage

- Matching algorithm
- Game suggestions

3.4.3 Prompts (Like Hinge)

Examples

- “The way to win me over is...”
- “My biggest green flag is...”

Purpose

- Humanize profile
- Improve chat starters

3.5 OPTIONAL EXTENDED PROFILE (POWER USERS)

3.5.1 Lifestyle Details

- Smoking / Drinking
- Workout frequency
- Diet (Veg/Non-veg)

3.5.2 Education & Career

- Education level
- Profession
- Company (optional)



3.5.3 Relationship Intent Details

- Casual / Serious / Marriage
- Open to long distance

3.5.4 Social Links (Optional)

- Instagram (verified)
- Spotify (music taste)

3.6 PROFILE COMPLETENESS ENGINE

Score Components

Component	Weight
Basic Info	20%
Photos	20%
Bio	10%
Interests	15%
Voice	20%
Quiz	15%

UX

- Progress bar
- “Complete your profile to get better matches”

3.7 BACKEND ARCHITECTURE

Tables

User Table

- user_id
- basic info

Profile Table

- bio
- interests
- preferences

Media Table

- photo URLs
- voice

Quiz Table

- answers
- personality score



Storage

- PostgreSQL (structured data)
- Redis (cache)
- S3 (media)

3.8 DATA PROCESSING & MATCHING INPUT

Data Used for Matching

- Intent
- Interests
- Personality
- Location
- Activity

Derived Metrics

- Compatibility score
- Engagement likelihood

3.9 EDGE CASE HANDLING

- No photo → restrict visibility
- Fake profile → flagged
- Incomplete → nudged
- Spam bio → filtered

3.10 PRIVACY & CONTROL (CRITICAL FOR QUIETCHAT)

Features

- Hide profile photos (optional mode)
- Blur images for non-matches
- Control who can view

User Controls

- Edit anytime
- Delete data
- Private mode

3.11 UX OPTIMIZATION STRATEGY

- Step-by-step onboarding
- Skip options (but incentivized completion)
- Smart autofill
- Micro rewards for completion

3.12 ENGAGEMENT IMPACT

- Better matches → higher retention
- More complete profiles → higher chat rate
- Voice + prompts → faster bonding



3.13 MONETIZATION IMPACT

- High-quality profiles → more calls
- Better matching → more spending
- Premium filters (future scope)

3.14 FUTURE ENHANCEMENTS

- AI profile suggestions
- Auto bio generator
- Smart photo ranking
- Personality AI matching

4. INTENT ENGINE

4.1 Purpose

Align user expectations early → reduce mismatch → improve retention.

4.2 Data Model

intent_type:

- dating
- casual
- serious
- friendship

4.3 Matching Impact

- Hard filter for incompatible intents
- Weighted scoring (40%)

4.4 UX Strategy

- Mandatory selection
- Can be updated later

5. USER VERIFICATION SYSTEM

5.1 Selfie Verification Flow

- Capture selfie
- Perform liveness (blink/head tilt)
- Compare with profile images

5.2 Backend

- Face recognition API (AWS Rekognition / Face++)
- Confidence threshold (>85%)

5.3 Badge Logic

- Verified users get ranking boost
- Increased trust visibility



5.4 Fraud Detection

- Duplicate faces across accounts
- Stock images detection

6. GAMIFIED ONBOARDING

6.1 This-or-That

- Binary questions (fast decision)
- Builds preference vectors

6.2 Personality Quiz

- 5–10 structured questions
- Outputs personality type

6.3 Voice Prompt

- Optional but rewarded
- Improves match success

6.4 Reward System

- Credits on completion
- Boost in profile score



7. PROFILE SCORING SYSTEM

7.1 Formula (System Logic)

Score = Completeness (30%) + Quiz (30%) + Voice (20%) + Activity (20%)

7.2 Dynamic Updates

- Score recalculated on activity
- Decay if inactive

7.3 Impact

- Match ranking priority
- Visibility boost

8. MATCHING ENGINE

8.1 Spin-Based Matching

- User spends token
- Weighted random selection

8.2 Algorithmic Matching

Inputs

- Intent
- Interests
- Distance
- Profile Score

Logic

Weighted scoring + filtering

8.3 Constraints

- Daily match limits
- Cooldown after excessive swipes

8.4 Edge Cases

- Low user density → expand radius
- No matches → fallback suggestions

9. MATCH FEED EXPERIENCE

9.1 UI Components

- Card layout
- Tags
- Online status

9.2 Actions



- Spin
- View profile

9.3 Behavioral Design

- Scarcity (limited spins)
- Urgency triggers

10. CHAT SYSTEM

10.1 Features

- Real-time messaging
- Media support (future)

10.2 Cost Logic

- 1 message = 1 credit
- Free messages for first interaction

10.3 Engagement Hooks

- Icebreakers
- Smart replies

10.4 Backend Architecture

- WebSocket connection
- Message persistence (DB)

10.5 Edge Cases

- Message spam → throttling
- Block/report system

11. AUDIO & VIDEO CALL SYSTEM

11.1 Audio Calls

- 1 credits/min
- Low latency

11.2 Video Calls

- 1 credits/min
- HD adaptive streaming

11.3 Controlled Video Call (Core Differentiator)

Description

A privacy-first video system where visual identity is revealed only after mutual consent.

Detailed Flow



1. User initiates video call
2. Call connects → both users appear blurred
3. 45 seconds interaction (voice + blurred visuals)
4. Consent prompt triggered for both users
5. Outcomes:
 - Both Accept → video unblurred
 - One Declines → remain blurred or continue audio

UX Components

- Blur overlay (Gaussian blur)
- Countdown timer (45 sec)
- Consent popup (Accept / Decline)

Backend Logic

- Sync consent state (both users must approve)
- Timer-based trigger event
- State machine:
 - INIT → BLURRED → CONSENT → CLEAR / CONTINUE

Safety Benefits

- Reduces harassment
- Encourages comfort
- Builds trust

Monetization Impact

- Longer call duration
- Increased call conversion
- Higher female retention

11.4 Billing Engine

- Per-second/min deduction
- Real-time wallet sync

11.5 Safety

- Report
- Auto-cut on low balance

12. WALLET & MONETIZATION

12.1 Wallet Structure

- Tokens for matches
- Minutes

12.2 Revenue Streams

- Subscriptions
- Top-ups
- Google Ads





12.3 Pricing Strategy (India)

- Low entry price
- Microtransactions

12.4 Payment Flow

- Select plan → Razorpay → success webhook → wallet update

13. USER LIFECYCLE

13.1 Core Loop

Login → Match → Chat → Call → Spend → Recharge

13.2 Funnel Optimization

- Reduce drop-off at onboarding
- Increase chat initiation

14. POST-MATCH MICRO GAMES (ENGAGEMENT ENGINE)

14.1 Objective

Increase engagement after match, reduce awkwardness, and accelerate bonding using ultra-short games.

14.2 Game Design Principles

- Duration: 10–45 seconds
- 1–2 interactions max
- No learning curve
- Emotion-driven
- Reward-based

14.3 Core Games

14.3.1 This or That – Match Battle

Concept: Choose between two options

Flow:

- 3 rounds
- Both answer → reveal

Outcome:

- Match % or winner

Impact:

- Starts conversation naturally

14.3.2 Fast Tap Challenge

Concept: Tap fastest in 5 seconds

Flow:

- Countdown → tap → result

Outcome:



- Highest taps wins

Impact:

- Competitive + fun

14.3.3 Guess the Number

Concept: Pick number 1–10

Flow:

- Both choose → reveal

Outcome:

- Closest wins

Impact:

- Quick + casual

14.3.4 Truth Pick (Safe Mode)

Concept: Answer light personal question

Outcome:

- No winner → bonding

Impact:

- Emotional connection

14.3.5 Emoji Match

Concept: Choose emoji

Outcome:

- Same = strong vibe

Impact:

- Expressive + relatable

14.4 Game Trigger System

Entry Points

- Chat → “🎮 Play Game”
- After match popup
- Push notifications

Smart Triggers

- No chat → suggest game
- Low engagement → trigger

14.5 Gameplay Flow

1. Invite sent
2. Match accepts
3. Game starts
4. Both complete action
5. Result calculated
6. Reward given
7. Continue chat



14.6 Reward System

Winner Rewards

- +5 Messages
- +1 Min Video Call
- +2 Min Audio Call

Rules

- Daily cap: 3
- Expiry: 24 hrs

14.7 Backend Architecture

Services

- Game Engine
- Sync Service
- Reward Engine

Data

Game Session:

- session_id
- game_type
- result

Reward:

- user_id
- reward
- expiry

14.8 Real-Time Logic

- WebSocket sync
- Async fallback

14.9 Abuse Prevention

- Max 5 games/day/match
- Reward limits
- Pattern detection

14.10 Engagement Impact

- Higher chat start rate
- Longer sessions
- Stronger bonding

14.11 Monetization Impact

- Increased feature usage
- Faster credit consumption
- Better retention